DepthWizard — Prototype Demo Video

SIH 2026 · PS **SIH26175** (ISRO / SAC) · Theme **Disaster Management**. Screen recording of the working prototype, narrated live. **Target 5–6 minutes.**

> **WHAT THIS VIDEO IS FOR** The deck already carries the problem, the architecture and the approach. So this video does not re-explain any of that. It does three things: it shows the prototype actually working on data it has never seen, it walks through what the tool can do, and it proves the accuracy by putting our output next to real airborne LiDAR. Explanation is kept to the single point that is genuinely novel, and even that gets thirty seconds.

Shape of the video

Time	On screen	What you are doing
0:00–0:30	Washington DC scene	What this is, in three sentences
0:30–1:05	Upload panel	Uploading Valkenburg live
1:05–1:35	Progress log	**The one novelty point**, while it runs
1:35–3:00	The new result	**Features.** Fly, layers, slope, measure, flood, rasters
3:00–4:10	Validation + toggle	**The proof.** Side by side against LiDAR
4:10–5:15	Accuracy tab	**The numbers.** This is the part to enjoy
5:15–5:45	Veluwe forest	One honest limitation, then close

Before you hit record

Step	Why
Start the server in **your own terminal**: `cd heimdall-web && npm run dev`	Then nothing else can stop it mid-take.
Check the GSD override box is empty	It only applies to images with no coordinate system. A value left in it overrides a GeoTIFF's own scale and puts a warning banner on screen.
Pre-run `valkenburg_hills` once, leave it in Recent jobs	Insurance. If the live run stalls you cut to a finished one with no dead air.
Open Washington DC, orbit once, then reload	Warms up the 3D so the first frame on camera is smooth.
Full screen, 100% zoom, notifications off	A popup mid-take costs a re-record.

> **Do not say a number you cannot show.** Every figure here is measured and sits in the repository. If a judge pauses the video and checks one, it holds up.

0:00 – 0:30 · What this is

> **Do:** Start with **Washington DC — civic** already open in the 3D flythrough. Orbit slowly while you talk.

SAY

This is DepthWizard, and everything you are looking at was reconstructed from a single ordinary photograph taken from above. No stereo pair, no LiDAR, no second pass over the same ground.

It is a real surface in real metres, so you can fly through it, measure distances on it, and ask it how high things are. Let me show you it working on an image it has never seen before.

0:30 – 1:05 · Upload it live

> **Do: New processing job** → drag `valkenburg_hills__UPLOAD_THIS.tif` in → confirm **GSD override is empty** → **Process image**.

SAY

This is a real aerial photograph of Valkenburg, a hill town in the south of the Netherlands. It is one of eight sites we deliberately kept out of training, so the model has never seen this place, and I am uploading it exactly as it comes with nothing tuned for this image. The two fields underneath are optional, for your own terrain model or control points, and I am leaving them empty.

1:05 – 1:35 · The one thing worth explaining

> **Do:** Let the pipeline stages light up. Point at them but keep talking.

SAY

While that runs, one point about why this works, because it is the part that surprised us.

The obvious approach is a depth model that outputs metres directly, and we tried that first. It fails, because those models assume a camera with perspective and a satellite image looking straight down has none. Its correlation with real LiDAR height was zero point zero six, which is noise. Predicting only relative depth and training a small network on LiDAR to convert it into metres gets you to zero point four one before any training at all. That measurement is the whole design.

1:35 – 3:00 · What the tool actually does

THIS IS THE LONGEST SECTION, AND IT SHOULD BE Nearly ninety seconds of showing features. Keep moving, give each one a sentence, and do not stop to justify anything.

Do: The job finishes and the **3D flythrough** opens. Orbit once, then switch to **Fly** and move along the hillside at roof height.

SAY

That is the surface, from the single photograph I uploaded about ten seconds ago. Every height is a real height in metres, with no vertical exaggeration anywhere in this view, and because Valkenburg is built on a hillside you can see the terrain and the buildings being treated as separate things.

Do: Click **Elevation**, pause. Click **Above ground**, pause. Click **Slope**, pause. Return to **Optical**.

SAY

The same reconstruction gives you three layers. Absolute elevation above sea level. Height above ground, with the terrain taken out. And slope, calculated directly from the surface, which is what you would use for access routes or landslide risk.

Do: Click **Measure**, draw a line across the slope over a couple of buildings, and let the elevation profile draw underneath.

SAY

You can measure on it, which gives distance along the ground with the height profile underneath, taken from the reconstruction rather than the flat photograph.

Do: Tick **Flood / water level**, drag the slider up several metres, hold, then untick.

SAY

And because this problem sits under disaster management, there is a water level tool. It cuts a plane through the scene at whatever height you choose and shows you what ends up underneath it. On a hill town it reads immediately, with the valley floor going under while the upper streets stay clear.

Do: Click the **2D maps** tab, scroll through the rasters, then return to **3D flythrough**.

SAY

And none of this is only a picture. Every run writes standard geospatial products as ordinary GeoTIFFs that open in QGIS with the coordinate system intact, so the 3D view is the interface but the rasters are the real deliverable.

3:00 – 4:10 · Proving it

Slow down here. Anybody can show a 3D render that looks plausible. Almost nobody puts the real measurement next to it and invites you to compare.

Do: Scroll to **Validate against reference**. Upload `valkenburg_hills__truth_heights.tif`. Wait for the metrics and the error map.

SAY

Now, you should not believe any of that just because it looks convincing. So I am handing the system the actual airborne LiDAR for this exact piece of ground and asking it to score itself against it. It reports the error three ways, and rather than reducing everything to one number it draws a map showing where the error actually is.

Do: The **Compare against the reference** toggle appears. Flip **Heimdall ↔ Reference** two or three times, slowly, holding the camera still.

SAY

And this is our reconstruction flipped back and forth against the real LiDAR. Same grid, same texture, same vertical origin, so the only difference between them is the heights themselves.

The hillside tracks, the street pattern tracks, and the open ground lands within about twenty centimetres. None of this was prepared in advance, because that is the file I dragged in at the start of the video.

Do: Open **Example scenes → Philadelphia — riverside**, toggle **Heimdall ↔ LiDAR truth** once, then leave it.

SAY

And the same comparison on an American city, using GAMUS, which is the dataset the problem statement recommends. So this is not one country, one sensor, or one kind of landscape.

4:10 – 5:15 · The numbers

> **THIS IS THE SECTION TO ENJOY** You have earned it by this point: they have watched it work and seen it checked against LiDAR. Deliver these figures with confidence, but keep the predict-zero comparison in — it is what makes them credible rather than boastful.

> **Do:** Click the **Accuracy** tab. Let it fill the frame. Point at the mean row, the landscape rows, then the resolution rows, slowly.

SAY

Every number behind that comparison is built into the application itself, rather than sitting on a slide.

Across eight held-out sites, scored against airborne LiDAR, we get five point three two metres RMSE on height above ground, at a correlation of zero point six seven. If you assumed the world was flat and predicted nothing, you would get eight point three seven. And the ground itself, which is most of the pixels in any image, comes in at eighty-two centimetres.

SAY

On the absolute surface model, terrain and structures together, which is what this problem is actually scored on, we get six point one one metres. About half a metre of that is not our model at all. It is a constant offset between the global terrain datum and the local one, and a single ground control point removes it, taking us to five point six three.

SAY

The problem statement asks for stability across urban, sparse, hilly and forested landscapes, so we measured each separately rather than hiding behind an average. Forest two point two four metres, sparse farmland one point five six, the hilly town I uploaded three point two one, and dense urban eight point four, which is the hardest case.

And since the evaluation uses Cartosat 2S at sixty centimetres, here is the same model with the imagery degraded to match. Five point three five at twenty-five centimetres. Five point five one at sixty. And five point seven two at one metre.

5:15 – 5:45 · One limitation, then close

> **Do:** Open **Example scenes** → **Veluwe — forest**. Orbit gently until you finish.

SAY

One limitation, stated plainly, because you will notice it if you look closely. Tall structures come out too short — on the tallest buildings we predict roughly half their true height, and tree canopy does the same, which is why I have put a forest on screen while I say it.

A photograph taken straight down contains no parallax, so when the model cannot tell a ten metre building from a twenty metre one, predicting something in between is mathematically correct. We could scale those up so the render looked better, and the measured error would get worse.

SAY

All of this runs on an ordinary laptop from a single command, with a CPU-only path if there is no GPU. The trained model is six and a half megabytes, and every figure I have quoted regenerates from a script in the repository.

One photograph goes in, and a measured three-dimensional surface you can explore comes out. Thank you for watching.

Delivery notes

- **Read it aloud once with a timer first.** Anywhere a sentence feels awkward in your mouth, change the words. Sounding like yourself matters more than matching this exactly.
- **The two moments that carry the video** are the reference toggle at 3:30 and the limitation at 5:15. Slow down for both. Volunteering a weakness reads as confidence.
- **Keep the features section moving.** One sentence per feature, no justifying. The point is breadth: it is a working tool, not a single trick.
- **Mouse discipline.** Hover a control for a beat before clicking so the viewer's eye arrives first. No fast spins in the 3D view.
- **If the live job fails while recording**, do not restart it. Say "I have one already processed here" and click it in **Recent jobs**.
- **Do not show** the terminal, the code, or your file system.
- **Running long?** Cut the Philadelphia comparison at 4:05, then the 2D maps beat at 2:45. Never cut the toggle or the limitation.
- **Running short?** Let the flythrough run in silence for a few seconds. Quiet over a moving 3D scene looks confident.

If they ask afterwards

Question	Answer
"How can one image give height at all?"	The scale is learned, not measured. The backbone gives relative ordering; a small head converts that to metres using patterns from sixteen thousand LiDAR-measured tiles — shadow, texture, apparent size, context. Where the image is ambiguous we under-predict, and we publish that bias.
"Five point three metres sounds like a lot."	It is dominated by tall structures. Ground and low vegetation are both under a metre. Predicting nothing scores 8.37.
"Have you tested on ISRO imagery?"	Not yet — we have no ISRO LiDAR to score against, and the FAQ confirms reference data will not be provided. We have tested resolution, and at the 0.6 m Cartosat 2S resolution we get 5.51 m.
"What about coarser than 2 m?"	It degrades and we say so. By 10 m a building spans one to three pixels. The pipeline warns when the input is outside its trained range.
"How do you know the improvements are real?"	Six controlled experiments changing one thing each, plus the same configuration run twice on different seeds to measure luck — about a tenth of a metre. Two of our earlier results turned out to be noise and we reported them as noise.
"Does it need a GPU?"	No. CPU fallback is built in; `run.sh` detects the hardware.

Every figure spoken, and its source

Figure	Value
Height above ground, 8 held-out LiDAR sites	**5.32 m** RMSE · r 0.67
Absolute DSM — what this problem is scored on	**6.11 m**, or 5.63 with one control point
Predicting zero height everywhere	8.37 m
Ground	0.82 m
Forest · farmland · hilly town · dense urban	2.24 · 1.56 · 3.21 · 8.40 m
0.25 m · **0.6 m (Cartosat 2S)** · 1 m	5.35 · **5.51** · 5.72 m
Metric versus relative backbone correlation	0.06 versus 0.41
Valkenburg, uploaded live	3.21 m RMSE · r 0.73
Model and speed	6.6 MB head · about 4.5 s per 400 m site

Regenerate with `python scripts/benchmark_ahn.py`. Figures read from `outputs/benchmark_ahn/results.json` and `outputs/evaluation_envelope.json`. Repository: github.com/Aryan241/Heimdall